

# Ahlan — Complete Cost Book

Every price, where it goes, measured scenarios & monthly projections

Measured from live production traces — 5 Aug 2026 · FX: 1 USD = 49 EGP

## 1 · The price sheet — every rate we pay, and to whom

These are the only external costs in the system. Everything else (routing, pricing, bills, automations) is code and costs nothing per use.

| What                 | Provider & model              | Rate                                   | Used for                                         |
|----------------------|-------------------------------|----------------------------------------|--------------------------------------------------|
| Chat brain (default) | OpenAI gpt-4.1-mini           | \$0.40 in / \$1.60 out per 1M tokens   | Casual replies, order extraction, phrasing       |
| Premium voice        | OpenAI gpt-4.1                | \$2.00 / \$8.00 per 1M                 | Angry guests, VIPs, first-timers (auto-escalate) |
| Router / choices     | OpenAI gpt-4.1-nano           | \$0.10 / \$0.40 per 1M                 | Message classification + option resolving        |
| Voice notes          | OpenAI 4o-mini-transcribe     | \$0.003 per minute (per second billed) | VN -> text; A/B'd better than whisper-1          |
| Photo understanding  | OpenAI gpt-4o-mini vision     | \$0.15 / \$0.60 per 1M                 | Guest sends a photo (complaints, menu asks)      |
| WhatsApp messages    | Meta                          | FREE inside 24h service window         | ALL current traffic incl. automations            |
| Template messages    | Meta (future, needs approval) | ~\$0.03–0.11 per message (Egypt)       | Win-backs, reorder reminders — off today         |
| Hosting              | Railway (flows + api)         | \$5–10 / month flat                    | The two backend services                         |
| Dashboard + DB       | Vercel + Supabase             | \$0 (free tiers, current scale)        | Dashboard hosting, database, storage             |

Where money does NOT go: no per-order commission, no per-message WhatsApp fee in the 24h window, no per-seat license, no charge for fast-path replies, status pushes, courier pushes, or the recovery/upsell/review automations (all code).

## 2 · Cost per message type (measured, last 7 days of real traffic)

| Message type                        | What happens inside                                                  | Cost (USD) | Cost (EGP) |
|-------------------------------------|----------------------------------------------------------------------|------------|------------|
| Fast-path answer                    | Pure code: hours, location, price, price math, thanks, approved FAQs | \$0.0000   | 0.00       |
| Router-only turn                    | nano classify + code dispatch                                        | \$0.0001   | 0.005      |
| Median chat/order turn              | mini extraction + code + mini phrasing                               | \$0.0012   | 0.06       |
| Average turn (all real traffic)     | mix of the above                                                     | \$0.0031   | 0.15       |
| Heavy chat turn (p90)               | long context or premium-model escalation                             | \$0.0103   | 0.50       |
| Voice-note add-on (7s)              | transcription only, then normal turn                                 | \$0.00035  | 0.017      |
| Photo from guest                    | vision classify + reply                                              | +\$0.0002  | +0.01      |
| Photo TO guest / menu PDF / receipt | sending media is free (in-window)                                    | \$0.0000   | 0.00       |
| Automation push                     | recovery / upsell / review — templated code                          | \$0.0000   | 0.00       |

### 3 · Real tested scenarios — ran today on production, costs from the traces

Each scenario below was actually executed through the live pipeline today. 'Turns' = guest messages. LLM cost is the trace-recorded total; VN cost added where voice was used.

| Scenario (as tested)                               | Turns | Tokens in/out | LLM   | +Voice | TOTAL EGP   |
|----------------------------------------------------|-------|---------------|-------|--------|-------------|
| "Total for 2 meals and a coke?" (price math)       | 1     | 0 / 0         | 0.000 | -      | <b>0.00</b> |
| Hours in Arabic / ingredients / thanks             | 1     | 0 / 0         | 0.000 | -      | <b>0.00</b> |
| "Send me a photo of the Iconic" (real image sent)  | 1     | 4.8k / 181    | 0.290 | -      | <b>0.29</b> |
| Delivery order, 3 turns, receipt PDF (O-C8GR)      | 3     | 5.7k / 312    | 0.242 | -      | <b>0.24</b> |
| Full VOICE order, 3 Arabic VNs (O-7J4F)            | 3     | 7.2k / 344    | 0.235 | 0.05   | <b>0.28</b> |
| Smart-edit order: absorb+imply+add+remove (O-HBKT) | 6     | 13.5k / 687   | 0.428 | -      | <b>0.43</b> |
| Classic pickup, options step by step (O-6P85)      | 7     | 15.9k / 686   | 0.445 | -      | <b>0.45</b> |
| Islam's real buggy VN session (6 VNs, repeats)     | 9     | 23.2k / 887   | 0.488 | 0.21   | <b>0.70</b> |
| Menu question with PDF attached                    | 1     | 4.8k / 192    | 0.501 | -      | <b>0.50</b> |
| Franco chit-chat (premium-voice escalation)        | 1     | 8.3k / 464    | 0.760 | -      | <b>0.76</b> |
| Injection attempt / bot-or-human / offers honesty  | 1     | ~4.5k / 190   | ~0.45 | -      | <b>0.45</b> |

Read: a COMPLETE order costs 0.24–0.45 EGP. Single chit-chat questions that miss every fast path cost 0.29–0.76 EGP because they carry the menu + persona context; they are the optimization frontier (more approved FAQs in Settings push them to 0.00). The buggy Islam session shows why bugs cost money: 3 wasted turns = +0.2 EGP. Those bugs are fixed.

## 4 · Scenario walkthroughs — the story, then the math

### A. Guest asks price then leaves

"bekam el loaded fries?" (Arabic) -> exact DB price. "shokran" -> closer.

- Turn 1: fast path (code) = 0.00
- Turn 2: closer (code) = 0.00

**Total: 0.00 EGP**

### B. Classic text pickup order (5 turns)

Item -> options in one block -> answers -> payment -> confirm. Receipt PDF attached.

- 2 order turns @ ~0.06 = 0.12
- options/confirm turns partly code = 0.10
- receipt + PDF send = 0.00

**Total: ~0.30 EGP**

### C. Full VOICE order — 3 Arabic voice notes (measured: O-7J4F)

One VN with everything -> pay VN -> confirm VN. New transcriber.

- 3 VNs x ~7s transcription = 0.05
- 3 LLM turns = 0.23

**Total: 0.28 EGP**

### D. Voice order with accents + one garbled word

"diblo" (misspelled Diablo) fuzzy-matched by code (free); truly unclear word -> ONE nano resolver call.

- Same as C = 0.28
- resolver (nano, only when needed) = 0.005

**Total: ~0.29 EGP**

### E. Guest sends a food photo + complaint

Vision reads the photo, bot empathizes, staff flagged if serious.

- vision = 0.01
- reply turn = 0.06-0.50 (mood-dependent)

**Total: 0.07-0.51 EGP**

### F. Abandoned order recovered

Guest builds a draft, goes silent 45 min. Code push with buttons; guest taps Yes; order placed.

- recovery push = 0.00
- "yes" confirm turn = 0.06
- receipt = 0.00

**Total: 0.06 EGP -> a recovered ORDER**

### G. Delivery day for one order

Order (3 turns) + courier auto-assigned + 4 driver-page pushes (on way / near / arrived / delivered) + review ask after 1h.

- order = 0.24
- assignment + 4 pushes + review ask = 0.00

**Total: 0.24 EGP for the entire lifecycle**

### H. Browse-only guest

"eh el burgerat 3andoko?" (Arabic) -> burgers listed; "photo?" -> image; leaves.

- category ask = 0.43 (LLM lists w/ menu context)
- photo = 0.29

**Total: ~0.72 EGP - the most expensive guest type; more FAQs => cheaper**

## 5 · Predictions — per customer, per restaurant, weekly & monthly

Assumptions (conservative): average order = 5 guest turns; 30% of orders fully voice; every order accompanied by 1.5 browse/chat conversations at ~0.30 EGP each; automations free; FX 49.

| Volume                                          | Orders cost | Chat/browse cost | AI total /mo           | Hosting     | ALL-IN /mo         | Per week |
|-------------------------------------------------|-------------|------------------|------------------------|-------------|--------------------|----------|
| 300 orders/mo                                   | 105 EGP     | 135 EGP          | 240 EGP                | 250-490 EGP | <b>490-730 EGP</b> | 115-170  |
| 1,000 orders/mo                                 | 350 EGP     | 450 EGP          | 800 EGP                | 250-490 EGP | <b>1,050-1,290</b> | 245-300  |
| 3,000 orders/mo                                 | 1,050 EGP   | 1,350 EGP        | 2,400 EGP              | 490 EGP     | <b>~2,890 EGP</b>  | ~670     |
| Per-customer view                               |             |                  | Value                  |             |                    |          |
| One complete order conversation                 |             |                  | <b>0.24 - 0.45 EGP</b> |             |                    |          |
| Typical customer / month (3 orders + 2 browses) |             |                  | <b>1.5 - 2.0 EGP</b>   |             |                    |          |

|                                                  |                  |
|--------------------------------------------------|------------------|
| Cost as % of a 300-EGP average ticket            | ~0.1 - 0.15%     |
| 1,000-order restaurant: AI cost per order all-in | ~1.05 - 1.29 EGP |

## 6 · Error margins & what could move these numbers

- Everything in section 3 is MEASURED, not estimated; sections 5's mix (turns/order, voice %, browse ratio) is the assumption layer — track it monthly from /api/metrics.
- Upside: every approved FAQ converts a paid chat turn to a free one forever; fast paths already answer common questions at 0.
- Downside risks: premium-voice escalations (angry guests) cost ~4x mini — capped by design to moods/VIPs; very long voice rambles (cap VNs at 60s is recommended); Meta template campaigns when enabled are a separate, per-send marketing budget.
- OpenAI price changes pass straight through; a model-tier switch (Settings -> AI) is one toggle.
- Multi-restaurant: each tenant pays only its own usage; hosting is shared — the marginal restaurant costs ONLY its AI usage (no new fixed cost until scale demands bigger instances).

## 7 · Revenue model — charging 1% per order

What Ahlan earns if the fee is 1% of each order's value. Anchors from the live Just Smash menu: cheapest realistic order (one sandwich) = 165 EGP -> 1.65 EGP fee; a single meal = 195-265 EGP -> ~2.00-2.65; measured average ticket ~300 EGP -> 3.00 per order.

| Order type                  | Ticket (EGP) | 1% fee     | Our marginal cost | Margin per order    |
|-----------------------------|--------------|------------|-------------------|---------------------|
| Minimum — one sandwich      | 165          | 1.65       | ~0.30-0.45        | ~1.2-1.35 (73-82%)  |
| One meal                    | 195-265      | 1.95-2.65  | ~0.30-0.45        | ~1.5-2.35 (77-83%)  |
| Average ticket (measured)   | ~300         | 3.00       | ~0.35-0.55        | ~2.45-2.65 (82-88%) |
| Family order / bundle (4X4) | 999-1,139    | 9.99-11.39 | ~0.45-0.70        | ~9.3-10.9 (93%+)    |

### Monthly picture per restaurant (revenue = 1% fee, costs = section 5 all-in):

| Volume                 | Revenue floor (all min orders) | Revenue expected (avg ticket) | All-in cost        | PROFIT (expected)    |
|------------------------|--------------------------------|-------------------------------|--------------------|----------------------|
| 300 orders/mo          | 495 EGP                        | 900 EGP                       | 490-730 EGP        | +170 to +410 EGP     |
| 1,000 orders/mo        | 1,650 EGP                      | 3,000 EGP                     | 1,050-1,290 EGP    | +1,710 to +1,950 EGP |
| 3,000 orders/mo        | 4,950 EGP                      | 9,000 EGP                     | ~2,890 EGP         | ~+6,100 EGP          |
| 10 restaurants x 1,000 | 16,500 EGP                     | 30,000 EGP                    | ~11,000-13,000 EGP | ~+17,000-19,000 EGP  |

Reading it: at 1% the fee on ONE average order (3 EGP) pays for ~7 orders' worth of AI. Break-even per restaurant is ~200-350 orders/month (where 1% covers the fixed hosting share); every order after that is ~82-88% margin. The floor row is the worst case where literally every order is a single cheapest sandwich. Voice orders reduce margin by only ~0.05 EGP each. Not included: Meta template campaigns (marketing, billed to the restaurant) and payment-gateway fees if online payment lands (typically borne by the restaurant's Paymob account).